Fikile Mkhize 0210210785080

0781087624

Core US

US 114636

DEMONSTRATE AN UNDERSTANDING OF PREVENTATIVE MAINTANANCE, ENVIRONMENTAL AND SAFETY ISSUES IN A COMPUTER ENVIRONMENT

FORMATIVE ASSESSMENT

SECTION A

QUESTION 1

1. Dust is a major killer of speed if you have not cleaned it up in over a year. The accumulation of dust blocks filters and prevent air from flowing through the computer.
2. Overheating is too much heat that can damage delicate circuitry. PC's components themselves are a cause of overheating, poorly ventilated environment, direct sunlight and dust.
3. Electrical disturbances may be caused by a problem that occurs on Eskom's transmission or distribution lines such as a lightning strike; it can also be caused by circumstances within your home or office building.
4. Thermal creep or chip creep is a PC problem that is caused by thermal expansion.

QUESTION 2

I. Ventilation- Ensure proper airflow in computer cases and rooms. Use fans, keep vents unblocked and avoid placing computers in enclosed spaces. This prevents overheating of CPU, GPU and other parts.

I. Uninterrupted Power Supply provides backup power during power outages and protects against surges or spikes. It gives time to safely shut down and prevents data loss and hardware damage.

II. Cleaning and dusting- Regularly clean keyboards, vents and internal components with compressed air and antistatic cloth. This prevents dust buildup that causes overheating and short circuits.

QUEESTION 3

I. 1. Class A fires involve ordinary combustible materials such as wood, paper or trash and are slow in their development and growth, and because these are solids they are easier to contain.

2. Class B fires involve flammable and combustible liquids and flammable gases, they develop and grow rapidly and because they are fluid they flow and move.

Fikile Mkhize 0210210785080

3. Class C fires involve energized electrical equipment such as motors. It is a live electrical circuit.

4. Class D fires involve combustible metals, rare in offices but possible in labs.

II. Install smoke detectors and fire alarms.

Keep Class C fires extinguishers near computer equipment.

Avoid overloading power outlets and extension cords.

Regular electrical inspections and proper cable management.

Have an evacuation plan and staff training.

QUESTION 4

I. 1. Electrical hazards- Exposed wires, overloaded sockets
2. Fire hazards- Overheating equipment, faulty wiring
3. Trip hazards- Loose cables on the floor
4.Ergonomic hazards- Poor posture and repetitive strain injury
5. Chemical hazards- Cleaning solvents and printer toner

II. 1. Electrical- Use surge protectors, don't overload sockets, change faulty equipment
2. Fire- Use correct extinguishers, no smoking and clear exits.
3. Trip- Use cable covers, route cables away from walkways.
4. Ergonomic- Adjust chairs or desks, take breaks and have proper monitor height.
5. Chemical- Store in labelled containers, use in ventilated areas and use PPE.

QUESTION 5

Electrostatic discharge is the transfer of charge between bodies at different electrical potentials. It is also describes as the electrical charge caused by an imbalance of electrons on the surface of the material.

Antistatic mat is a grounding mat such as a floor or table mat used to reduce the risk of electrostatic discharge.

Wrist strap can also be used to prevent electrostatic shock , you wrap it around the wrist and has a cord that attaches to the computer.

Touch a grounded metal surface before touching components.

Maintain humidity levels to reduce static buildup.

Fikile Mkhize 0210210785080

QUESTION 6

Disposing of batteries

1. Remove the batteries from the computer and place it in a recyclable bag or other closed container.
2. Contact your local municipal authority or private trash collector to inquire about the policies for hazardous waste removal.
3. Check out local retail stores that sell computer equipment, they provide recycle bins.

Disposing of CRTs

Dispose a Cathode Ray Tube at a hazardous-waste recycling center.

-Avoid landfill dumping ,it prevents toxic chemicals from leaching into soil and water.

-Data destruction; wipe or physically destroy hard drives before disposal to prevent data leaks.